

Prove trigonometric identities



1 Prove the following identities:

a $\frac{\sin x}{\cos x \tan x} = 1$

c $\sin A \cos A \tan A = \sin^2(A)$

e $\frac{\sin^2(x)}{\cos^2(x)} = \tan^2(x)$

b $\frac{\cos x \tan x}{\sin x} = 1$

d $\frac{\sin x \cos(90^\circ - x)}{\cos x \sin(90^\circ - x)} = \tan^2(x)$

2 Prove the following identities:

a $(\sin x + \cos x)^2 = 1 + 2 \sin x \cos x$

c $2 \cos^2(\theta) - 3 = -1 - 2 \sin^2(\theta)$

e $(1 + \tan^2(\alpha)) \cos^2(\alpha) = 1$

b $\cos^4(x) - \sin^4(x) = 2 \cos^2(x) - 1$

d $5 \cos^2(\theta) - 3 = 2 - 5 \sin^2(\theta)$

f $\tan^2(y) - \sin^2(y) = \tan^2(y) \sin^2(y)$

3 Prove the following identity:

$$\sin^2(a) - \sin^2(b) + \cos^2(a) \sin^2(b) - \sin^2(a) \cos^2(b) = 0$$

4 Prove the following identities:

a $\frac{\sin x \cos x}{\tan x} = \cos^2(x)$

c $\frac{1 - \sin^2(x)}{\cos x} = \cos x$

e $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$

g $\frac{(1 + \sin \theta)^2 + \cos^2(\theta)}{1 + \sin \theta} = 2$

b $\frac{1 - \cos^2(\theta)}{\sin \theta} = \sin \theta$

d $\frac{1 - \sin^2(x)}{\cos x} = \cos x$

f $\frac{\tan^2(x) - 1}{\tan^2(x) + 1} = 1 - 2 \cos^2(x)$

h $\frac{\sin^2(x) + \sin x \cos x}{\cos^2(x) + \sin x \cos x} = \tan x$

5 Prove the following identities:

a $\frac{\sec \theta}{\tan \theta} = \csc \theta$

c $\frac{\cos \theta}{1 - \sin \theta} - \tan \theta = \sec \theta$

e $\tan \theta \sin \theta + \cos \theta = \sec \theta$

g $\frac{4 + \tan^2(x) - \sec^2(x)}{\csc^2(x)} = 3 \sin^2(x)$

b $\frac{1 - \cot x}{1 + \cot x} = \frac{\tan x - 1}{\tan x + 1}$

d $\frac{\cos \alpha}{1 + \sin \alpha} = \sec \alpha (1 - \sin \alpha)$

f $\cot \theta (\sec^2(\theta) - 1) = \tan \theta$

6 Prove the following identity:



$$\sin \theta (1 + \tan \theta) + \cos \theta (1 + \cot \theta) = \frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta}$$